INTEGRATED EMPIRICAL, ENGINEERING AND POLICY WORKFLOW FRAMEWORK

Observed Registry Microdata, Semi-Empirical Naval Resistance, Spatial Diagnostics, Frontier Efficiency and Conditional Policy Scenarios

1A. Observed Fleet Register

- Subarea 29 vessel register microdata
- 6,584 active Turkish vessels across 34 ports
- 675.7 megawatts and 87,364 gross tons
- Purse seiners, trawlers, dredgers, small boats
- Mean age 22.6 years, 9.99 percent over 35

1B. Observed Catch Accounts

- National Black Sea catch 256.7 thousand tons
- Area 37 regional context 1950 to 2024
- Area 37 cumulative 41.9 million tons
- Pelagic, demersal and mollusc species guilds
- Anchovy, sprat, bonito and sea snail

1C. Observed Cost Accounts

- National marine fishery cost structure 2023
- Fuel represents 20.70 percent of revenue
- Crew share labor equals 19.00 percent
- Gross cash operating margin is 44.52 percent
- Depreciated net margin is 29.52 percent

1D. Macro and Ocean Series

- Central Bank annual exchange rate series
- Consumer price index inflation deflators
- Basin sea surface temperature anomalies
- Satellite surface chlorophyll series
- Illustrative satellite speed check subset

Vessel Dimensions and Power

Port Registry and Catch

Cost Shares and Guilds

Macro and Ocean Series

PHASE 2: Semi-Empirical Naval Hydrodynamics

- Fleet mean engine intensity is 14.53 kilowatts per ton
- Gear-specific displacement and bollard benchmarks
- 43.9 percent of trawlers exceed 10 kilowatts per ton
- Holtrop-Mennen and frictional resistance decomposition
- Transit wave drag reaches 58 to 79 percent of resistance
- Active towing wave drag remains under 1.3 percent
- Displacement-normalized power checks confirm ranking
- Within-gear scaling slopes remain negative for large vessels

PHASE 3: Spatial Port Autocorrelation Analysis

- Full 34-port audit: 27 Black Sea plus 7 Marmara ports
- Four-nearest-neighbor coastal weight matrix
- Port engine intensity Moran index is 0.641
- Dual local spatial cluster testing with false discovery:
  - High-intensity western shelf trawling cluster
  - Low-intensity eastern pelagic purse-seine zone
- Aggregate port power shows dispersed hub structure
- Priority industrial hubs identified for shore power

PHASE 4: Cohort Frontier Eco-Efficiency Model

- Four-step exclusion audit yields 4,772 Black Sea vessels
- 21 basin, gear and length operational cohorts
- Inputs: certified engine power and gross tonnage
- Desirable outputs: pelagic, demersal and mollusc catch
- Undesirable output: 305.6 thousand tons carbon dioxide
- Directional distance frontier with constant-returns check
- Single-output dimensionality and allocation robustness
- Technical slack equals 30.4 thousand tons per year

Transit and Towing Regimes

Port Intensity Clusters

Frontier Abatement Slack

Basin Catch and Fleet Series

PHASE 5: Duty-Cycle Carbon Inventory and Policy Scenarios

- Gear-differentiated operational baseline: 367.0 thousand tons
- Uniform industrial capacity upper bound: 784.9 thousand tons
  - Scenario A: 441-kilowatt limit across 315 industrial vessels
  - Scenario B: Retirement of 658 vessels over 35 years old
  - Scenario C: Shift of 165 trawl vessels to passive gear
  - Scenario D: 15-percent technical propulsion efficiency gain
  - Scenario E: Integrated limit and retirement with 42-vessel overlap
- Surviving-fleet Monte Carlo simulation with 10,000 draws:
  - Contrasts zero catch rebound against five-percent relief
- Fuel shock stress tests across 8 to 20 percent depreciation
- Integrated abatement: 107.6 thousand tons or 29.3 percent

PHASE 6: Multi-Basin Panel Portfolio Volatility Analysis

- Five Turkish marine basins observed from 2000 to 2024
- 105 five-year rolling windows plus 25 non-overlapping blocks
- Dependent variables: revenue and catch variation coefficients
- Basin fixed effects with autocorrelation-robust standard errors
- Higher catch species diversity lowers volatility at modest scale
- Positive diversity-by-fleet interaction shows scale erosion
- Mean-centered diversity slope evaluated at sample mean fleet
- Exchange rate depreciation amplifies revenue instability
- Consumer price inflation and thermal anomaly controls included
- Non-overlapping block estimation confirms negative diversity slope
- Fleet rightsizing restores ecological revenue stabilization

Conditional Scenarios A to E

Industrial Hub Shore Power

Scale and Diversity Evidence

PHASE 7: Conditional Decarbonization Policy Roadmap and Fisheries Governance

- 1. Certified Engine Power Standards: Phase in a 441-kilowatt main engine limit for trawlers and transitional limits for large seiners
  - Pair continuous shaft-power monitoring with high-thrust nozzles and propeller repitching to maintain low-speed bollard pull
- 2. Targeted Older-Vessel Decommissioning: Retire 658 vessels exceeding 35 years of service alongside strict license-transfer bans
  - Integrated power limits and retirements cut realistic annual emissions by 107.6 thousand tons while raising surviving margins
- 3. Selective Shore-Side Electrification: Deploy cold-ironing connections at high-power industrial ports led by Samsun, Istanbul and Trabzon
  - Target auxiliary generator loads of large purse seiners and refrigerated trawlers during peak seasonal landing queues
- 4. Reformed Fuel Tax Incentives: Redirect unconditional diesel excise exemptions toward verified propulsion and hull upgrades
  - Link modernization co-financing to audited specific fuel oil consumption reductions and low-drag gear adoption
- 5. Multilateral Black Sea Coordination: Harmonize Subarea 29 capacity rules, catch documentation and monitoring across coastal states
  - Shared regional effort discipline prevents cross-border fishing displacement on migratory anchovy and sprat stocks